

MAXIM: Metasurfaces-oriented electromagnetic wave simulation software with intuitive graphical user interfaces^{☆,☆☆}

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ARTICLE INFO

Article history:

Received 6 May 2020

Received in revised form 17 November 2020

Accepted 18 December 2020

Available online 10 March 2021

Keywords:

Computational electromagnetics

Rigorous coupled-wave analysis

Scattering matrix method

Metasurface

Graphical user interface

Portable software

ABSTRACT

We develop MAXIM which is electromagnetic wave simulation software based on rigorous coupled-wave analysis. The principal advantage of MAXIM is an intuitive graphical user interface drastically improving the accessibility of the software to who are not familiar with computer programming. Here, we present the basic formulation and computation methods that are incorporated in MAXIM. The computation performance is also evaluated for several didactic examples of dielectric metasurfaces which are the main application of MAXIM. The comparison of the calculation results with commercial software based on a finite-difference time-domain method confirms that the computation results of two programs coincide closely with each other within 1% difference. Considering the easy accessibility, wide availability and high reliability, MAXIM will serve the development of related research fields of metasurfaces and nanophotonics.

Program summary

Program Title: MAXIM

CPC Library link to program files: <https://doi.org/10.17632/352jpd593h.1>

Licensing provisions: LGPL

Programming language: Python 3.8

Supplementary material: User guide and tutorial movie

Nature of problem: Time-harmonic electromagnetic wave simulations on multilayered periodic structures composed of isotropic materials. Diffraction occurs as a propagating electromagnetic wave meets periodically-structured arrays of materials that have different refractive indices. Wave parameters such as amplitude and phase of diffracted waves are modulated depending on the structure configuration. The main purpose of this software is to calculate complex transmission and reflection coefficients of diffracted waves from dielectric metasurfaces that are composed of arrays of subwavelength antennas.

Solution method: Rigorous coupled-wave analysis associated with an extended scattering matrix method. According to Bloch's theorem, diffracted waves from periodic structures can be represented as a truncated Fourier series in which the primitive reciprocal vector is the same as that of periodic structures. Appropriate boundary conditions at the interface of a single structure layer enable development of an eigenvalue equation for which the solution gives a scattering matrix of the layer, and the coupling coefficients within it. In the case of a multilayered structure, a total scattering matrix can be obtained using the Redheffer star product, which interconnects scattering matrices of each layer. Total coupling coefficients can be also calculated using the extended Redheffer star product; this ability is a major advantage of the extended scattering matrix method. Diffraction from periodic structures can be fully described by the total scattering matrix and total coupling coefficients.

Additional comments including restrictions and unusual features: MAXIM builds on several open-source Python packages including PySide2 [1], Numpy [2], Scipy [3], Pandas [4], Matplotlib [5] and Pyinstaller [6].

References

[1] PySide2, https://wiki.qt.io/Qt_for_Python

[☆] The review of this paper was arranged by Prof. Stephan Fritzsche.

^{☆☆} This paper and its associated computer program are available via the Computer Physics Communication homepage on ScienceDirect (<http://www.sciencedirect.com/science/journal/00104655>).

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- [2] Numpy, <https://numpy.org/>
- [3] Scipy, <https://www.scipy.org/>
- [4] Pandas, <https://pandas.pydata.org/>
- [5] Matplotlib, <https://matplotlib.org/>
- [6] Pyinstaller, <https://www.pyinstaller.org/index.html>

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1. Introduction

Metasurfaces that consist of subwavelength optical antenna arrays have shown exceptional capabilities as ultrathin flat optical devices [1]. Conventional optical components such as lenses control light by exploiting refraction, so the final devices are usually bulky and heavy. In contrast, metasurfaces in which thickness is comparable to the operating wavelength can perform the same functions because tiny antennas of metasurfaces can efficiently interact with incident light. Due to those advantages, many promising applications have been demonstrated using metasurfaces such as ultrathin lenses [2], holograms [3–5], structural colors [6–8] and skin cloaks [9].

Metasurfaces can be classified as plasmonic or dielectric, according to the nature of the material [10]. **Plasmonic metasurfaces exploit surface plasmons, which are collective electron oscillations at the interface between conductors and dielectrics.** Reflective plasmonic metasurfaces of metal–insulator–metal configurations exploit Fabry–Perot cavity effects, and therefore can achieve very high modulation efficiency in the visible spectrum [11]; however, the modulation efficiency of transmissive plasmonic metasurfaces is usually very low compared with conventional refractive optical elements.

Dielectric metasurfaces have high modulation efficiency for both reflective and transmissive types. Scattering properties of dielectric antennas are determined by multipole resonances that incident light induces inside antennas; the resonances can be tailored by modifying the geometry of antennas [12]. Many types of antennas have been developed to control wave parameters such as amplitude, phase and frequency. Simultaneous control of multiple parameters is also possible [13,14]. Recently efforts are focused on active tunability [15] as well as largescale manufacturing of dielectric metasurfaces for practical applications [16,17].

Appropriate design of metasurfaces requires understanding of transmission or reflection characteristics of individual antennas. For this purpose, several computational electromagnetic (EM) wave simulation methods can be used, such as the finite element method (FEM) and finite-difference time-domain method (FDTD). Such methods are well developed and benefit from high degrees of freedom (DoF) in simulation, including boundary conditions, light sources and structure configurations.

Rigorous coupled-wave analysis (RCWA) is an appropriate method to efficiently calculate the EM responses of metasurfaces [18]. RCWA has distinct advantages over other methods. **First, RWCA can provide analytic solutions of diffracted waves, whereas other methods, in principle, provide only numerical electric or magnetic field data. Therefore, RCWA can easily discriminate among diffracted waves by their diffraction orders. In RWCA, the calculation time is not affected by the thickness of structures, so nanostructures combined with multiple thick layers can be considered without difficulty. In particular cases such as one dimensional (1D) gratings on a distributed Bragg reflector, RCWA shows outstanding computation speed and accuracy. Several open-source packages based on RCWA were already developed to boost the advance of nanophotonics [18–22].** However, those scripting-based programs suffer from

limited structural configuration or low accessibility to users who are not familiar with computer programming. There are some open-source RCWA programs with a graphical user interface (GUI), but they must be incorporated in internet connection or other paid software [23,24].

Here, we develop MAXIM which is open-source RCWA software to calculate the EM response of periodically distributed structures composed of isotropic materials, with the purpose of guiding design of dielectric metasurfaces. MAXIM features a powerful GUI that improves the accessibility of the software to users who do not have expertise in programming languages. MAXIM has four distinct advantages as follows.

- **Highly-intuitive GUI.** All simulation variables are tabulated in the main table widget, so users can check their simulation conditions easily. Most open-source RCWA codes require users to manually define each variable and corresponding values; this process usually causes troubles because users may not be aware of all necessary variables and available options in a comprehensive simulation environment. In contrast, MAXIM displays all the required variables in the table, so the whole simulation can be understood without difficulty.
- **High-DoF in structure configurations.** MAXIM can consider various kinds of structures including films, rectangles, circles, arbitrary polygons and their combinations. Truncated cones and rectangular horns composed of multiple alternating materials are also available. These objects are essential in the characterization of a side-wall effect, which is one of the most common fabrication defects in dielectric metasurfaces.
- **Convenient parametric sweep.** When designing metasurfaces, a parametric sweep is unavoidable because general analytic scattering solutions of non-spherical objects rarely exist. MAXIM contains powerful functions to build a calculation schedule for the parametric sweep. Thus, users can easily construct a large simulation schedule as desired.
- **Portable software.** The package of MAXIM incorporates an executable file that can be run in a Windows operating system. MAXIM does not require preloaded packages, compilation processes, or virtual environments, so beginners have real opportunities to access RCWA.

The calculation accuracy of MAXIM is also verified by comparing its calculation results with those of commercial FDTD simulation software (Lumerical). RCWA solves the Maxwell equations in a frequency domain whereas FDTD computes electric and magnetic fields in a time domain. Due to the obvious contrast between two frameworks, FDTD is suitable to evaluate the calculation accuracy of RCWA. We calculate complex transmission and reflection coefficients from several exemplary structures such as Huygens, geometric and chiral metasurfaces. **The calculation results of MAXIM agree within 1% with those of Lumerical.** Moreover, MAXIM automatically provides the complex coefficients after calculation, so additional computation is not necessary; this attribute is highly convenient for the process of designing metasurfaces.

This article presents the basic formulation and computation methods of MAXIM. Section 2 presents a brief overview of RCWA formulation implemented in MAXIM. Section 3 introduces the computation methods of MAXIM such as the data structures, logic and internal processes of particular functions. Section 4 validates the computation performance of MAXIM through several didactic examples. Section 5 summarizes the capability and competence of MAXIM.

2. Basic formulation

Since the 1960s, coupled-wave analysis that is a predecessor of current RCWA has been developed to study sinusoidal diffraction gratings [25,26], and the original RCWA formulation is founded in 1995 [27]. RCWA calculates diffraction from multilayered periodic structures by solving algebraic Maxwell equations in a spatial frequency domain. Bloch's theorem provides the fundamental basis of RCWA, so EM waves in each layer are expanded to a truncated Fourier series of Bloch eigenmodes that propagate in the normal direction of the layers. EM responses such as transmission, reflection and field information can be obtained from Bloch eigenmodes, so the objective of RCWA is to specify the complex amplitude, wavevectors and coupling coefficients of every Bloch eigenmode of each layer.

In this section, we summarize the RCWA formulation implemented in MAXIM. The basic formulation begins with the time-harmonic Maxwell equations. It can be converted into an algebraic form by substituting Bloch eigenmode expressions for electric and magnetic fields. In the algebraic Maxwell equations, dielectric functions are also expanded into a truncated Fourier series, and can be represented by Toeplitz matrices for 1D gratings, and as block Toeplitz matrices with Toeplitz blocks (BTBs) for two dimensional (2D) gratings [28]. Then the eigenvalue equation for wavevector components in a propagating direction can be derived by eliminating longitudinal field components. Most of the computation costs of RCWA are consumed by the process of solving the eigenvalue equations, and the cost increases rapidly as the truncation order increases. The solutions of the eigenvalue equation give a complete description of Bloch eigenmodes within a single layer. Therefore, transmission and reflection properties of the layer can be computed by implementation of appropriate boundary conditions on inplane field components.

In a multilayered system, the characteristics of each layer can be used to determine the total properties of the system by interconnecting the layers. Several matrix techniques can be used to achieve this interconnection. The classical formulation uses a non-recursive algorithm of a transmission matrix; this method has the benefits of low computation cost, but has suffered from low numerical stability for lossy materials [27]. The enhanced transmittance matrix approach can improve the stability of the method [29]. Recursive matrix algorithms such as the scattering matrix (S-matrix), R-matrix and hybrid matrix methods can increase the numerical stability by suppressing exponential growth of sub-matrices during computation [30]. The S-matrix method is perhaps a dominant approach among the recursive algorithms because of higher calculation efficiency and accessibility than other recursive methods. Various kinds of S-matrix methods with different details have been developed to study nanophotonics and to improve the capability of RCWA [31–35]. Each of the methods has their own specialties. The basic formulation of MAXIM is equivalent to conventional RCWA formulation. In addition, MAXIM exploits the extended S-matrix method (ESMM) to calculate total EM responses of the multilayered system by connecting each layer [34]. The core of the ESMM is the extended Redheffer star product allowing the calculation of the coupling coefficients of Bloch eigenmodes in a manner to the use of the Redheffer star product for the S-matrix. Therefore, the ESMM provides an opportunity for parallel computation of EM fields.

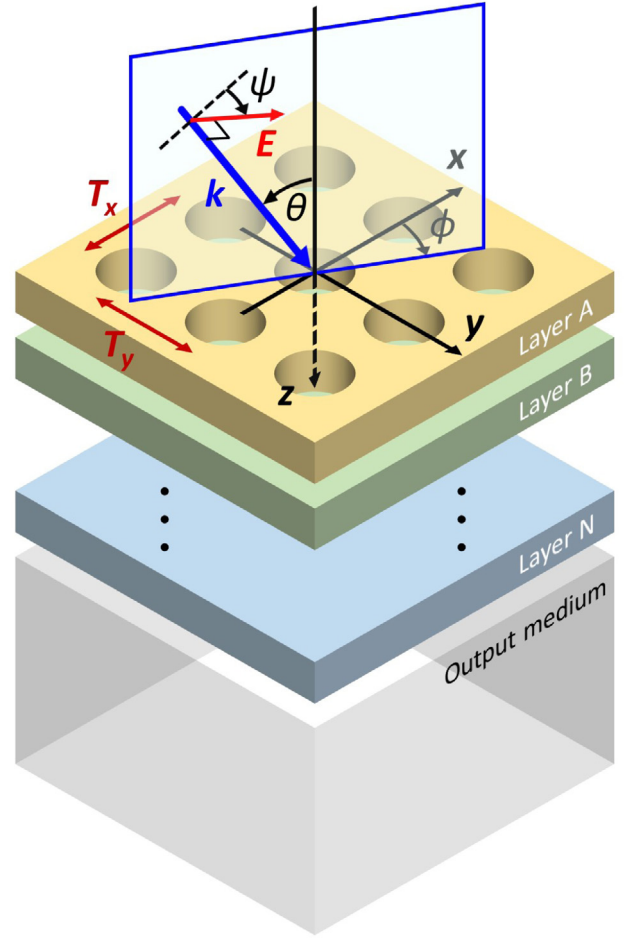


Fig. 1. The coordinate system of MAXIM. An electromagnetic plane wave is incident from the input medium and transmitted to the output medium through intermediate layers. E : electric field vector; k : wavevector; T_i : spatial period along each direction. The geometrical properties of the incidence are represented by θ , ϕ and ψ .

2.1. Rigorous coupled-wave analysis

To simplify understanding, the terminology of the subsequent considerations complies with the matrix expressions of the precedent works [34–37] due to their intelligible language. Our coordinate system also conforms with the traditional system in which each structure layer is positioned at the xy plane invariant in the z direction (Fig. 1). EM waves propagate in the $+z$ direction.

For an isotropic homogeneous nonmagnetic medium without any sources, time-harmonic Maxwell curl equations with time dependence of $e^{-j\omega t}$ can be represented as [38]

$$\nabla \times \vec{E} = j\omega\mu_0\mu\vec{H} \quad (1)$$

$$\nabla \times \vec{H} = -j\omega\epsilon_0\epsilon\vec{E} \quad (2)$$

where $\vec{E} = (E_x, E_y, E_z)$ is the electric field, $\vec{H} = (H_x, H_y, H_z)$ is the magnetic field, ω is the angular frequency, ϵ_0 and μ_0 indicate the permittivity and permeability of vacuum, respectively. Relative permittivity ϵ and permeability μ of the medium are spatially-varying periodic functions, so they can be approximated by the truncated Fourier series as

$$\epsilon = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \epsilon_{m,n} e^{jG_{m,n} \cdot \mathbf{r}} \quad (3)$$

$$\mu = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \mu_{m,n} e^{j\mathbf{G}_{m,n} \cdot \mathbf{r}} \quad (4)$$

where $\varepsilon_{m,n}$ is a Fourier expansion coefficient of permittivity and $\mu_{m,n}$ is a Fourier expansion coefficient of permeability, $\mathbf{G}_{m,n} = (2\pi m/T_x, 2\pi n/T_y)$ represents the inplane reciprocal lattice vector, T_x and T_y denote spatial periods, $\mathbf{r}$ is the inplane position vector, M is the number of Fourier harmonics (also called a truncation order) along the x -direction, N is the number of Fourier harmonics along the y -direction. The truncation order of permittivity and permeability are set to $2M$ and $2N$ for convenience of calculation. Bloch's theorem yields the forms of electric $\vec{\mathbf{E}}$ and magnetic $\vec{\mathbf{H}}$ fields as

$$\vec{\mathbf{E}} = \sum_{m=-M}^M \sum_{n=-N}^N \vec{\mathbf{E}}_{m,n} e^{j[(\mathbf{k}_{xy} + \mathbf{G}_{m,n}) \cdot \mathbf{r} + \kappa z]} \quad (5)$$

$$\vec{\mathbf{H}} = j\sqrt{\frac{\varepsilon_0}{\mu_0}} \sum_{m=-M}^M \sum_{n=-N}^N \vec{\mathbf{H}}_{m,n} e^{j[(\mathbf{k}_{xy} + \mathbf{G}_{m,n}) \cdot \mathbf{r} + \kappa z]} \quad (6)$$

where $\vec{\mathbf{E}}_{m,n}$ and $\vec{\mathbf{H}}_{m,n}$ represent the complex amplitude vectors of the electric and magnetic fields respectively of each Bloch eigenmode, and $\mathbf{k}_{xy} = (k_x, k_y)$ is the discrete inplane wavevector determined from Bloch's theorem. For convenience of calculation, the magnetic fields of Eq. (6) are normalized by the factor of $j\sqrt{\varepsilon_0/\mu_0}$, which is the free space impedance. The calculation accuracy is usually increased by increasing M and N , but the computational cost also grows rapidly. The longitudinal wavevector component κ which determines whether each Bloch eigenmode becomes a propagation wave or evanescent wave can be obtained by solving the succeeding eigenvalue equation.

By substituting Eqs. (3), (4), (5), (6) into Eqs. (1) and (2), the algebraic form of the Maxwell equations can be derived and reduced into the matrix form of [35]

$$jk_0 \begin{bmatrix} \llbracket \varepsilon \rrbracket & 0 & 0 \\ 0 & \llbracket \varepsilon \rrbracket & 0 \\ 0 & 0 & \llbracket \varepsilon \rrbracket \end{bmatrix} \begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_y \\ \mathbf{e}_z \end{bmatrix} = \begin{bmatrix} 0 & -\mathbf{K}_z & \mathbf{K}_y \\ \mathbf{K}_z & 0 & -\mathbf{K}_x \\ -\mathbf{K}_y & \mathbf{K}_x & 0 \end{bmatrix} \begin{bmatrix} \mathbf{h}_x \\ \mathbf{h}_y \\ \mathbf{h}_z \end{bmatrix} \quad (7)$$

$$jk_0 \begin{bmatrix} \llbracket \mu \rrbracket & 0 & 0 \\ 0 & \llbracket \mu \rrbracket & 0 \\ 0 & 0 & \llbracket \mu \rrbracket \end{bmatrix} \begin{bmatrix} \mathbf{h}_x \\ \mathbf{h}_y \\ \mathbf{h}_z \end{bmatrix} = \begin{bmatrix} 0 & -\mathbf{K}_z & \mathbf{K}_y \\ \mathbf{K}_z & 0 & -\mathbf{K}_x \\ -\mathbf{K}_y & \mathbf{K}_x & 0 \end{bmatrix} \begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_y \\ \mathbf{e}_z \end{bmatrix} \quad (8)$$

where k_0 is a wavenumber in vacuum, $\llbracket \varepsilon \rrbracket$ and $\llbracket \mu \rrbracket$ are BTBs formed by harmonic coefficients $\varepsilon_{m,n}$ and $\mu_{m,n}$ respectively performing 2D convolutions in Fourier space, $\mathbf{K}_i$ is a diagonal matrix formed by wavevector components along each direction, $\mathbf{e}_i$ and $\mathbf{h}_i$ are column vectors formed by complex amplitude components $\vec{\mathbf{E}}_{m,n}$ and $\vec{\mathbf{H}}_{m,n}$ respectively of Bloch eigenmodes along each direction. The sorting scheme of $\mathbf{K}_i$ is the same for that of the BTB. The additional rearrangement of Eqs. (7) and (8) produces the following eigenvalue equation of

$$\mathbf{P}\mathbf{Q} \begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_y \end{bmatrix} = \left(\frac{j\kappa}{k_0}\right)^2 \begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_y \end{bmatrix} \quad (9)$$

where

$$\mathbf{P} = \begin{bmatrix} -\mathbf{K}_x \llbracket \varepsilon \rrbracket^{-1} \mathbf{K}_y & -\llbracket \mu \rrbracket + \mathbf{K}_x \llbracket \varepsilon \rrbracket^{-1} \mathbf{K}_x \\ \llbracket \mu \rrbracket - \mathbf{K}_y \llbracket \varepsilon \rrbracket^{-1} \mathbf{K}_y & \mathbf{K}_y \llbracket \varepsilon \rrbracket^{-1} \mathbf{K}_x \end{bmatrix} \quad (10)$$

$$\mathbf{Q} = \begin{bmatrix} -\mathbf{K}_x \llbracket \mu \rrbracket^{-1} \mathbf{K}_y & -\llbracket \varepsilon \rrbracket + \mathbf{K}_x \llbracket \mu \rrbracket^{-1} \mathbf{K}_x \\ \llbracket \varepsilon \rrbracket - \mathbf{K}_y \llbracket \mu \rrbracket^{-1} \mathbf{K}_y & \mathbf{K}_y \llbracket \mu \rrbracket^{-1} \mathbf{K}_x \end{bmatrix} \quad (11)$$

and $^{-1}$ denotes the matrix inverse operation. From Eq. (9), the column vector $\mathbf{u}$ of the eigenvalues κ , and the corresponding eigenvector matrix $\mathbf{D}$ consisting of Fourier coefficients (complex amplitudes) $\vec{\mathbf{E}}_{m,n}$ and $\vec{\mathbf{H}}_{m,n}$ of a single layer can be determined. In subsequent formulation, $\vec{\mathbf{H}}_{m,n}$ is renormalized by $-j\sqrt{\mu_0/\varepsilon_0}$ for convenience of calculation. Furthermore, because of the square operator of κ in Eq. (9), the solution always includes a pair of waves; one propagating forward and one backward.

2.2. S-matrix method

The S-matrix of a single layer can be found using eigenvalues and the eigenvector of the layer when it is placed in vacuum. This assumption is plausible because the total S-matrix of multiple layers is calculated by exploiting infinitesimally thin buffer layers of vacuum; thus, the effect of the buffer layer becomes negligible. When the EM plane wave is incident to a single layer in a forward direction, then incidence, reflection and transmission waves can be represented as a Fourier series as

$$\begin{bmatrix} \vec{\mathbf{E}}^i \\ \vec{\mathbf{H}}^i \end{bmatrix} = \sum_{m=-M}^M \sum_{n=-N}^N \begin{bmatrix} \vec{\mathbf{E}}_{m,n}^i \\ \vec{\mathbf{H}}_{m,n}^i \end{bmatrix} e^{j[(\mathbf{k}_{xy} + \mathbf{G}_{m,n}) \cdot \mathbf{r} + k_z(z-z^-)]} \quad (12)$$

$$\begin{bmatrix} \vec{\mathbf{E}}^r \\ \vec{\mathbf{H}}^r \end{bmatrix} = \sum_{m=-M}^M \sum_{n=-N}^N \begin{bmatrix} \vec{\mathbf{E}}_{m,n}^r \\ \vec{\mathbf{H}}_{m,n}^r \end{bmatrix} e^{j[(\mathbf{k}_{xy} + \mathbf{G}_{m,n}) \cdot \mathbf{r} - k_z(z-z^-)]} \quad (13)$$

$$\begin{bmatrix} \vec{\mathbf{E}}^t \\ \vec{\mathbf{H}}^t \end{bmatrix} = \sum_{m=-M}^M \sum_{n=-N}^N \begin{bmatrix} \vec{\mathbf{E}}_{m,n}^t \\ \vec{\mathbf{H}}_{m,n}^t \end{bmatrix} e^{j[(\mathbf{k}_{xy} + \mathbf{G}_{m,n}) \cdot \mathbf{r} + k_z(z-z^+)]} \quad (14)$$

respectively, where $k_z = \sqrt{k_0^2 - k_x^2 - k_y^2}$ represents the z -direction wavenumber, z^- and z^+ are the lower and upper bounds of the layer, respectively (Fig. 2). The corresponding electric $\vec{\mathbf{E}}^g$ and magnetic $\vec{\mathbf{H}}^g$ fields inside the layer are

$$\begin{bmatrix} \vec{\mathbf{E}}^g \\ \vec{\mathbf{H}}^g \end{bmatrix} = \sum_{m=-M}^M \sum_{n=-N}^N \left\{ c_f^+ \begin{bmatrix} \vec{\mathbf{E}}_{m,n}^{g+} \\ \vec{\mathbf{H}}_{m,n}^{g+} \end{bmatrix} e^{j[(\mathbf{k}_{xy} + \mathbf{G}_{m,n}) \cdot \mathbf{r} + \kappa^+(z-z^-)]} + c_f^- \begin{bmatrix} \vec{\mathbf{E}}_{m,n}^{g-} \\ \vec{\mathbf{H}}_{m,n}^{g-} \end{bmatrix} e^{j[(\mathbf{k}_{xy} + \mathbf{G}_{m,n}) \cdot \mathbf{r} + \kappa^-(z-z^+)]} \right\} \quad (15)$$

where g denotes the space within the specific layer, c_f are the coupling coefficients of forward incidence denoted by c_f , and $^+$ and $^-$ represent the forward and backward propagation directions of Bloch eigenmodes, respectively. Tangential components of both electric and magnetic fields should be continuous at interfaces of $z = z^-$ and $z = z^+$, so Eqs. (12), (13), (14), (15) can be reduced to [37]

$$\begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V} & -\mathbf{V} \end{bmatrix} \begin{bmatrix} \vec{\mathbf{E}}^i \\ \vec{\mathbf{E}}^r \end{bmatrix} = \begin{bmatrix} \mathbf{W}^+(z^-) & \mathbf{W}^-(z^-) \\ \mathbf{V}^+(z^-) & \mathbf{V}^-(z^-) \end{bmatrix} \begin{bmatrix} \mathbf{C}_f^+ \\ \mathbf{C}_f^- \end{bmatrix} \quad (16)$$

at $z = z^-$

$$\begin{bmatrix} \mathbf{W}^+(z^+) & \mathbf{W}^-(z^+) \\ \mathbf{V}^+(z^+) & \mathbf{V}^-(z^+) \end{bmatrix} \begin{bmatrix} \mathbf{C}_f^+ \\ \mathbf{C}_f^- \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V} & -\mathbf{V} \end{bmatrix} \begin{bmatrix} \vec{\mathbf{E}}^t \\ 0 \end{bmatrix} \quad (17)$$

at $z = z^+$

where $\mathbf{I}$ is an identity matrix, $\mathbf{V}$ is a transformation matrix from electric fields to magnetic fields, $\mathbf{C}_f^+$ and $\mathbf{C}_f^-$ are column vectors formed by the coupling coefficient c_f^+ and c_f^- , respectively. Other matrices are given by

$$\mathbf{W}^+(z) = \mathbf{D}\mathbf{U}^+ \quad (18)$$

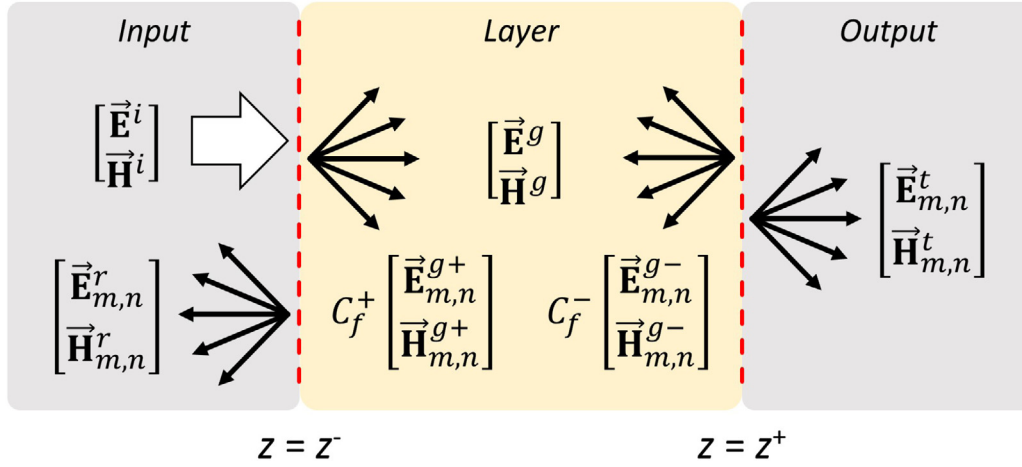


Fig. 2. Schematic of electromagnetic waves inside and outside the layer. The diverging arrows stand for the Bloch eigenmode expansion. The S-matrix of the single layer can be obtained from the boundary conditions at $z = z^+$ and $z = z^-$.

$$\mathbf{V}^+(z) = \frac{1}{k_0} \mathbf{P}^{-1} \mathbf{D} \mathbf{U}^+ \quad (19)$$

$$\mathbf{W}^-(z) = \mathbf{D} \mathbf{U}^- \quad (20)$$

$$\mathbf{V}^-(z) = \frac{1}{k_0} \mathbf{P}^{-1} \mathbf{D} \mathbf{U}^- \quad (21)$$

where $\mathbf{U}^+$ and $\mathbf{U}^-$ are diagonal matrices in which (i, i) th element is given by $e^{iu_i(z-z^-)}$ and $e^{iu_i(z-z^+)}$ respectively, u_i represents i th element of $\mathbf{u}$. Therefore, if $\vec{\mathbf{E}}^i$ is given, $\vec{\mathbf{E}}^r$, $\vec{\mathbf{E}}^t$, $\mathbf{C}_f^+$ and $\mathbf{C}_f^-$ can be obtained by solving Eqs. (16) and (17).

When $\vec{\mathbf{E}}^i = \mathbf{I}$, a forward transmission coefficient $\mathbf{T}_f$ and a forward reflection coefficient $\mathbf{R}_f$ are defined by $\vec{\mathbf{E}}^t$ and $\vec{\mathbf{E}}^r$, respectively. By replacing $\vec{\mathbf{E}}^i$ with $\mathbf{I}$ in Eqs. (16) and (17), a forward coupling coefficient matrix $\mathbf{C}_f$ is given by [37]

$$\begin{aligned} \mathbf{C}_f &= \begin{bmatrix} \mathbf{C}_f^+ \\ \mathbf{C}_f^- \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{W}^+(0) + \mathbf{V}^{-1} \mathbf{V}^+(0) & \mathbf{W}^-(z^-) + \mathbf{V}^{-1} \mathbf{V}^-(z^-) \\ \mathbf{W}^+(z^+) - \mathbf{V}^{-1} \mathbf{V}^+(z^+) & \mathbf{W}^-(0) - \mathbf{V}^{-1} \mathbf{V}^-(0) \end{bmatrix}^{-1} \\ &\quad \times \begin{bmatrix} 2\mathbf{I} \\ 0 \end{bmatrix} \end{aligned} \quad (22)$$

and corresponding $\mathbf{T}_f$ and $\mathbf{R}_f$ are represented by

$$\mathbf{T}_f = \mathbf{W}^+(z^+) \mathbf{C}_f^+ + \mathbf{W}^-(0) \mathbf{C}_f^- \quad (23)$$

$$\mathbf{R}_f = \mathbf{W}^+(0) \mathbf{C}_f^+ + \mathbf{W}^-(z^-) \mathbf{C}_f^- - \mathbf{I} \quad (24)$$

In a development similar to the steps from Eq. (12) to (24), a backward coupling coefficient matrix $\mathbf{C}_b$, transmission coefficient $\mathbf{T}_b$ and reflection coefficient $\mathbf{R}_b$ can be found as [37]

$$\begin{aligned} \mathbf{C}_b &= \begin{bmatrix} \mathbf{C}_b^+ \\ \mathbf{C}_b^- \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{W}^+(0) + \mathbf{V}^{-1} \mathbf{V}^+(0) & \mathbf{W}^-(z^-) + \mathbf{V}^{-1} \mathbf{V}^-(z^-) \\ \mathbf{W}^+(z^+) - \mathbf{V}^{-1} \mathbf{V}^+(z^+) & \mathbf{W}^-(0) - \mathbf{V}^{-1} \mathbf{V}^-(0) \end{bmatrix}^{-1} \\ &\quad \times \begin{bmatrix} 0 \\ 2\mathbf{I} \end{bmatrix} \end{aligned} \quad (25)$$

and

$$\mathbf{T}_b = \mathbf{W}^+(0) \mathbf{C}_b^+ + \mathbf{W}^-(z^-) \mathbf{C}_b^- \quad (26)$$

$$\mathbf{R}_b = \mathbf{W}^+(z^+) \mathbf{C}_b^+ + \mathbf{W}^-(0) \mathbf{C}_b^- - \mathbf{I} \quad (27)$$

where b represents the backward incidence. Finally, the S-matrix $\mathbf{S}$ of the single layer can be found as

$$\mathbf{S} \equiv \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{T}_f & \mathbf{R}_b \\ \mathbf{R}_f & \mathbf{T}_b \end{bmatrix} \quad (28)$$

2.3. Extended S-matrix method

In the case of multilayered structures, the total S-matrix can be found by connecting S-matrices of individual single layers. This operation denoted by the symbol $\otimes$ which is called the Redheffer star product. It assumes buffer layers of vacuum among each layer, so the total transmission and reflection properties are computed by considering infinite number of reflections within buffer layers. Then, the total S-matrix can be found in the limit of zero thickness of the buffer layers to remove phase retardation.

When two layers of A and B are connected, the total S-matrix $\mathbf{S}^T$ of the two layers can be represented as [34]

$$\begin{aligned} \mathbf{S}^T &= \mathbf{S}^A \otimes \mathbf{S}^B \\ &= \begin{bmatrix} \mathbf{T}_f^A & \mathbf{R}_b^A \\ \mathbf{R}_f^A & \mathbf{T}_b^A \end{bmatrix} \otimes \begin{bmatrix} \mathbf{T}_f^B & \mathbf{R}_b^B \\ \mathbf{R}_f^B & \mathbf{T}_b^B \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{T}_f^B (\mathbf{I} - \mathbf{R}_b^A \mathbf{R}_f^B)^{-1} \mathbf{T}_f^A & \mathbf{R}_b^B + \mathbf{T}_f^B (\mathbf{I} - \mathbf{R}_b^A \mathbf{R}_f^B)^{-1} \mathbf{R}_b^A \mathbf{T}_b^B \\ \mathbf{R}_f^A + \mathbf{T}_b^A (\mathbf{I} - \mathbf{R}_f^B \mathbf{R}_b^A)^{-1} \mathbf{R}_f^B \mathbf{T}_f^A & \mathbf{T}_b^B (\mathbf{I} - \mathbf{R}_f^B \mathbf{R}_b^A)^{-1} \mathbf{T}_b^A \end{bmatrix} \end{aligned} \quad (29)$$

where A and B specify the layers. Similarly, the total coupling coefficient matrix $\mathbf{C}^T$ can be also calculated by using the extended Redheffer star product $\otimes$ as [34]

$$\begin{aligned} \mathbf{C}^T &= \begin{bmatrix} \mathbf{C}_f^{A,T} & \mathbf{C}_f^{B,T} \\ \mathbf{C}_b^{A,T} & \mathbf{C}_b^{B,T} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{C}_f^A \\ \mathbf{C}_b^A \end{bmatrix} \otimes \begin{bmatrix} \mathbf{C}_f^B \\ \mathbf{C}_b^B \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{C}_f^A + \mathbf{C}_b^A (\mathbf{I} - \mathbf{R}_f^B \mathbf{R}_b^A)^{-1} \mathbf{R}_f^B \mathbf{T}_f^A & \mathbf{C}_f^B (\mathbf{I} - \mathbf{R}_b^A \mathbf{R}_f^B)^{-1} \mathbf{T}_f^A \\ \mathbf{C}_b^A (\mathbf{I} - \mathbf{R}_f^B \mathbf{R}_b^A)^{-1} \mathbf{T}_b^A & \mathbf{C}_b^B + \mathbf{C}_f^B (\mathbf{I} - \mathbf{R}_b^A \mathbf{R}_f^B)^{-1} \mathbf{R}_b^A \mathbf{T}_b^B \end{bmatrix} \end{aligned} \quad (30)$$

where elements of $\mathbf{C}^T$ represent updated coupling coefficient matrices. For multilayered structures, corresponding $\mathbf{S}^T$ and $\mathbf{C}^T$ can be obtained in a similar manner as

$$\mathbf{S}^T = \mathbf{S}^A \otimes \mathbf{S}^B \otimes \mathbf{S}^C \otimes \dots \otimes \mathbf{S}^N \quad (31)$$

$$\mathbf{C}^T = \begin{bmatrix} \mathbf{C}_f^{A,T} & \mathbf{C}_f^{B,T} & \mathbf{C}_f^{C,T} & \dots & \mathbf{C}_f^{N,T} \\ \mathbf{C}_b^{A,T} & \mathbf{C}_b^{B,T} & \mathbf{C}_b^{C,T} & \dots & \mathbf{C}_b^{N,T} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_f^A \\ \mathbf{C}_b^A \end{bmatrix} \otimes \begin{bmatrix} \mathbf{C}_f^B \\ \mathbf{C}_b^B \end{bmatrix} \otimes \begin{bmatrix} \mathbf{C}_f^C \\ \mathbf{C}_b^C \end{bmatrix} \otimes \dots \otimes \begin{bmatrix} \mathbf{C}_f^N \\ \mathbf{C}_b^N \end{bmatrix} \quad (32)$$

where N indicates the last layer. The size of $\mathbf{C}^T$ increases as the number of layers increases whereas the size of $\mathbf{S}^T$ is independent of the number of layers.

The remaining step is to connect input and output media into a multilayered structure in which EM response is described by Eqs. (31) and (32). Transmission and reflection characteristics of the semi-infinite input medium can be also found by considering boundary conditions of tangential field components at the interface. The boundary equations are described as

$$\begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V}^{\text{IN}} & \mathbf{V}^{\text{IN}} \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{R}_f^{\text{IN}} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V} & -\mathbf{V} \end{bmatrix} \begin{bmatrix} \mathbf{T}_f^{\text{IN}} \\ \mathbf{0} \end{bmatrix} \quad (33)$$

$$\begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V}^{\text{IN}} & \mathbf{V}^{\text{IN}} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \mathbf{T}_b^{\text{IN}} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V} & -\mathbf{V} \end{bmatrix} \begin{bmatrix} \mathbf{R}_b^{\text{IN}} \\ \mathbf{I} \end{bmatrix} \quad (34)$$

where IN indicates input medium and $\mathbf{V}^{\text{IN}}$ is a transformation matrix from electric fields to magnetic fields in the input medium. By solving Eqs. (33) and (34), the boundary S-matrix of input

medium $\mathbf{S}^{\text{IN}} = \begin{bmatrix} \mathbf{T}_f^{\text{IN}} & \mathbf{R}_b^{\text{IN}} \\ \mathbf{R}_f^{\text{IN}} & \mathbf{T}_b^{\text{IN}} \end{bmatrix}$ can be determined. In a similar manner to that used for the input medium, the boundary S-matrix of output medium $\mathbf{S}^{\text{OUT}} = \begin{bmatrix} \mathbf{T}_f^{\text{OUT}} & \mathbf{R}_b^{\text{OUT}} \\ \mathbf{R}_f^{\text{OUT}} & \mathbf{T}_b^{\text{OUT}} \end{bmatrix}$ can be also computed from the following boundary equations of output medium described as

$$\begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V} & -\mathbf{V} \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{R}_f^{\text{OUT}} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V}^{\text{OUT}} & \mathbf{V}^{\text{OUT}} \end{bmatrix} \begin{bmatrix} \mathbf{T}_f^{\text{OUT}} \\ \mathbf{0} \end{bmatrix} \quad (35)$$

$$\begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V} & -\mathbf{V} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \mathbf{T}_b^{\text{OUT}} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{V}^{\text{OUT}} & \mathbf{V}^{\text{OUT}} \end{bmatrix} \begin{bmatrix} \mathbf{R}_b^{\text{OUT}} \\ \mathbf{I} \end{bmatrix} \quad (36)$$

where OUT denotes output medium and $\mathbf{V}^{\text{OUT}}$ is a transformation matrix from electric fields to magnetic fields in the output medium. The final total S-matrix $\mathbb{S}$ of the multilayered structure combined with input and output media becomes

$$\mathbb{S} = \mathbf{S}^{\text{IN}} \otimes \mathbf{S}^T \otimes \mathbf{S}^{\text{OUT}} \quad (37)$$

and the final total coupling coefficient matrix $\mathbf{C}$ can be also found by updating all elements of $\mathbf{C}^T$ with $\mathbf{S}^{\text{IN}}$ and $\mathbf{S}^{\text{OUT}}$ according to Eq. (30).

The calculation results so far give a complete description of the diffraction from multilayered periodic structures because each Bloch eigenmode of transmission and reflection waves corresponds to actual diffracted light. Explicit formulas of $\bar{\mathbf{E}}^t$ and $\bar{\mathbf{E}}^r$ can be obtained by multiplying $\mathbb{S}$ and complex field amplitudes of a given $\bar{\mathbf{E}}^i$. Some straightforward manipulation of these formulas allows investigating diffraction characteristics such as diffraction powers, angles and complex field profiles. For example, the relative diffraction power $\eta_{m,n}$ for a particular diffraction order (m, n) is represented as

$$\eta_{m,n} = |\bar{\mathbf{E}}_{m,n}|^2 \frac{1}{k_{z,0}} \Re \left[\sqrt{k^2 - |\mathbf{k}_{xy} + \mathbf{G}_{m,n}|^2} \right] \quad (38)$$

where k is a wavenumber of input or output media, $k_{z,0}$ is a zeroth order wavevector component in the longitudinal direction, and the operator $\Re$ represents the real part of the complex number. Thus, $\eta_{m,n}$ becomes zero for sufficiently high orders while total transmittance and reflectance can be found by the sum of each

$\eta_{m,n}$. Complete electric and magnetic field profiles can be also recovered using Eqs. (12), (13), (14) and (15) consisting of $\bar{\mathbf{E}}_{m,n}$, $\bar{\mathbf{H}}_{m,n}$, $\mathbf{u}$, $\mathbf{D}$ and $\mathbf{C}$, which are already calculated above.

3. Computation methods

This section presents specific computation methods that use the formulation presented in the preceding section. A table is a fundamental apparatus of MAXIM. Simulation conditions are entered in the form of a table by users, and this table is converted to another form of a table by internal algorithms. Only the 'string' datatype is allowed in the former table, so its contents should be transformed to calculable forms of numbers in the latter table. This conversion process may look inefficient, but the input datatype of string can greatly improve the availability of user scripting in sweep functions to build enormous schedules.

The major advantage of the table is that users can easily recognize and manage their simulation conditions, so the accessibility of the software can be improved drastically. In MAXIM, columns of the table contain simulation conditions such as a wavelength, period and structure configuration while rows of the table represent different simulations, *i.e.*, schedules. At first, the table consists of environment conditions without structures, and users can build their own structures using various buttons in the toolbar. Simulations can be executed simply by completing the table.

MAXIM has two operation modes: a Normal mode, and a Sweep mode. In Normal mode, one simulation condition can be examined by a single execution to provide EM field profiles. In Sweep mode, many simulation conditions can be computed by a single execution; this mode is favorable for structure optimization.

3.1. Packages

MAXIM is written by Python (version 3.8) with other open-source packages to improve calculation convenience and build intuitive GUIs. Representative packages used in MAXIM and their functions are described as follows.

- PySide2: This package enables construction of GUIs that have many widgets. All GUIs of MAXIM are generated using PySide2.
- Numpy: This package provides linear algebraic functions for matrix operations, and many mathematical functions to calculate equations of the preceding section.
- Scipy: This package is very similar to Numpy, and both of them provide the function to eigenvalue equations. MAXIM uses Scipy to solve the eigenvalue problem of Eq. (9) because the calculation stability of Scipy is superior to that of Numpy for large matrices.
- Pandas: This package provides a new datatype of the table that is called Dataframe, and related functions to manipulate it. The schedule table written by users is converted to Dataframe by internal algorithms because Dataframe can simultaneously contain various forms of data such as numbers and text.
- Matplotlib: This package is exploited to draw plots of data such as refractive indices of materials, transmittance (reflectance) results, EM field profiles, and diffraction powers with corresponding angles.

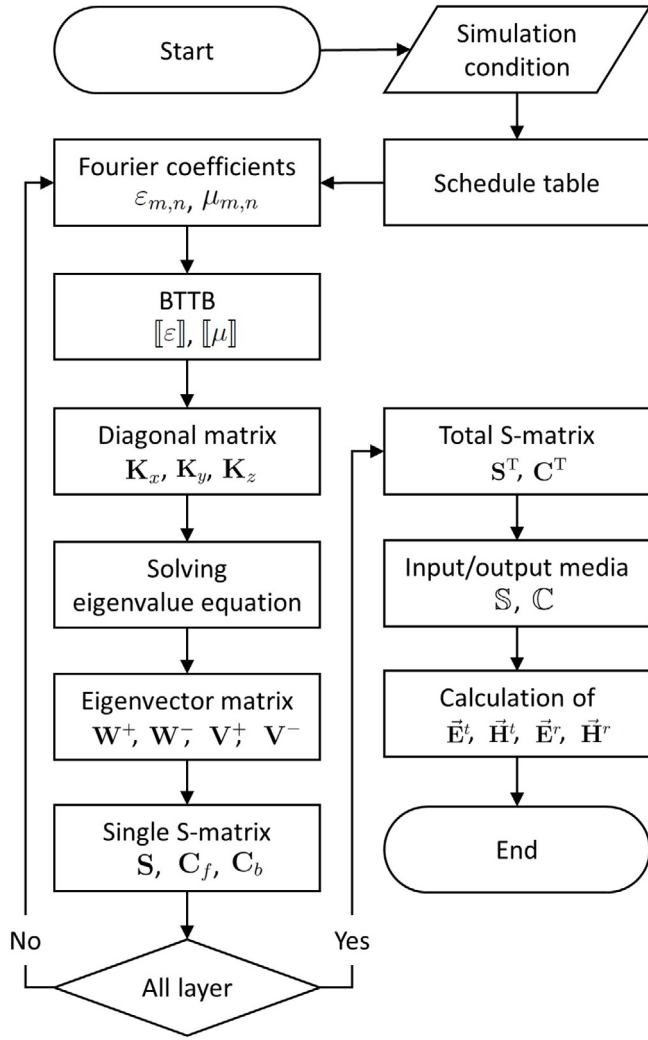


Fig. 3. A simplified flowchart representing the computation sequence of MAXIM.

3.2. Logic

The basic computation sequence of MAXIM follows the formulation described in the preceding section (Fig. 3). The table written by users is converted into the calculable table of Dataframe by internal algorithms; then the simulation proceeds row by row. First, the permittivity distribution of the structure layer is expanded as a Fourier series, and yields Fourier coefficients $\varepsilon_{m,n}$ of Eqs. (3). MAXIM assumes that all structures have permeability $\mu = 1$. Then, the BTTB $[[\varepsilon], [\mu]]$ is set up to build and solve the eigenvalue equation of Eq. (9). Then we obtain $\mathbf{u}$ and $\mathbf{D}$ which can be used to find $\mathbf{S}$, $\mathbf{C}_f$ and $\mathbf{C}_b$ of the single layer by calculating Eqs. (22) to (28). For single-layered structures, these variables directly work as $\mathbf{S}^T$ and $\mathbf{C}^T$ of Eqs. (31) and (32). For multilayered structures, the preceding process is repeated to obtain $\mathbf{S}$, $\mathbf{C}_f$ and $\mathbf{C}_b$ of each layer. Then, $\mathbf{S}^T$ and $\mathbf{C}^T$ are calculated by Redheffer operations of Eqs. (31) and (32). Finally, $\mathbf{S}$ and $\mathbf{C}$ are derived by incorporating $\mathbf{S}^{\text{IN}}$ and $\mathbf{S}^{\text{OUT}}$. Complex transmission and reflection coefficients by diffraction orders are easily obtained by multiplying $\mathbf{S}$ and the incident field vector that is defined by input simulation conditions.

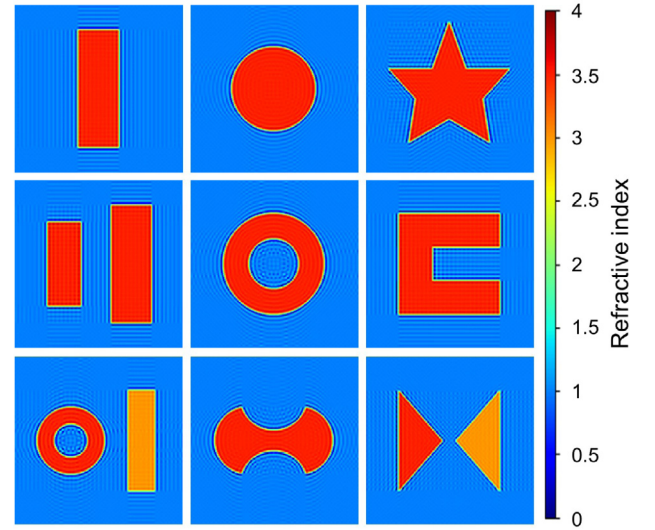


Fig. 4. Various index distributions in the xy plane for exemplary structures which can be generated in MAXIM.

3.3. Block Toeplitz matrix with Toeplitz blocks

One of the major advantages of MAXIM is high DoF in available structure configurations. The Fourier transform of ε is represented as a matrix $\tilde{\varepsilon}$ in which elements are given by $\varepsilon_{m,n}$, and $\tilde{\varepsilon}$ has the dimension of $(4M+1) \times (4N+1)$ regardless of structure types. Due to the assumption that $\mu = 1$, the Fourier transform of μ is always a constant matrix. MAXIM provides several functions to compute $\tilde{\varepsilon}$ of various kinds of structures such as films, rectangles, circles, arbitrary polygons and their combinations (Fig. 4). These structures are representative structures that have been widely investigated in metasurface research. Truncated cones and rectangular horns can be created by stacking rectangles or circles; for user convenience, MAXIM also provides simple functions to do this automatically. Once we have $\tilde{\varepsilon}$ of a certain structure, its EM responses can be characterized by following the formulation.

A film represents a homogeneous medium that can be defined by its thickness and refractive index. If the refractive index of a film is given by $\tilde{n}$, the Fourier coefficient matrix $\tilde{\varepsilon}$ of the film is defined as a matrix in which all elements are zero except the center element given by $\tilde{n}^2$. If no structure is provided, MAXIM automatically generates a zero-thickness film that has $\tilde{n} = 1$. This structure does not affect the calculation result. Thus, the simulation condition is the same as the single infinite interface between two different media.

A rectangular structure is defined by a length, width, center position, and the refractive indices of the structure and of its surrounding medium. The Fourier transform of a 2D rectangular function can be represented analytically using sinc functions, so (m, n) th component of $\tilde{\varepsilon}$ of the rectangular structure can be written as

$$\varepsilon_{m,n} = \begin{cases} (\tilde{n}_r^2 - \tilde{n}_s^2) \frac{w_x w_y}{T_x T_y} \text{sinc}\left(\frac{w_x m}{T_x}\right) \text{sinc}\left(\frac{w_y n}{T_y}\right) e^{-j\mathbf{G}_{m,n} \cdot \mathbf{d}} & \text{if } m \neq 0 \text{ or } n \neq 0 \\ \tilde{n}_s^2 + (\tilde{n}_r^2 - \tilde{n}_s^2) \frac{w_x w_y}{T_x T_y} & \text{if } m = 0 \text{ and } n = 0 \end{cases} \quad (39)$$

where $\tilde{n}_r$ and $\tilde{n}_s$ are the refractive indices of the rectangular structure and of its surrounding medium, w_x and w_y are length and width of the structure respectively, $\mathbf{d} = (d_x, d_y)$ is an inplane

position vector of the center, and d_i represents the coordinate of the center in each direction.

The Fourier coefficient matrices of other structures such as circles, arbitrary polygons and their combinations can be approximated using numerical methods. The internal algorithm of MAXIM generates a matrix of the image that corresponds to given structures, then the matrix is converted to $\tilde{\varepsilon}$ by a fast Fourier transform. In this way, arbitrary pillar structures invariant in the longitudinal direction can be simulated in MAXIM. Particular structures of truncated cones and rectangular horns can be assembled by stacking multiple circles or rectangles that have varying cross sections in the longitudinal direction. MAXIM also provides simple functions to generate them, and the related internal calculation processes are the same as for manually-stacked structures. The simulations of these structures are required for metasurface research because dielectric nanostructures that are fabricated using dry etching always have tilted side walls.

After the calculation of $\tilde{\varepsilon}$ of given structures, the BTTB $[\varepsilon]$ of Eq. (7) can be defined as follows. While conventional Toeplitz matrix $\mathbf{A}$ is defined by $A_{m,n} = A_{m+1,n+1}$, $[\varepsilon]$ has the form [39]

$$[\varepsilon] = \begin{bmatrix} [\varepsilon_0] & [\varepsilon_{-1}] & [\varepsilon_{-2}] & \cdots & [\varepsilon_{-2M}] \\ [\varepsilon_1] & [\varepsilon_0] & [\varepsilon_{-1}] & \cdots & [\varepsilon_{-2M+1}] \\ [\varepsilon_2] & [\varepsilon_1] & [\varepsilon_0] & \cdots & [\varepsilon_{-2M+2}] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ [\varepsilon_{2M}] & [\varepsilon_{2M-1}] & [\varepsilon_{2M-2}] & \cdots & [\varepsilon_0] \end{bmatrix} \quad (40)$$

where

$$[\varepsilon_m] = \begin{bmatrix} \varepsilon_{m,0} & \varepsilon_{m,-1} & \varepsilon_{m,-2} & \cdots & \varepsilon_{m,-2N} \\ \varepsilon_{m,1} & \varepsilon_{m,0} & \varepsilon_{m,-1} & \cdots & \varepsilon_{m,-2N+1} \\ \varepsilon_{m,2} & \varepsilon_{m,1} & \varepsilon_{m,0} & \cdots & \varepsilon_{m,-2N+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \varepsilon_{m,2N} & \varepsilon_{m,2N-1} & \varepsilon_{m,2N-2} & \cdots & \varepsilon_{m,0} \end{bmatrix} \quad (41)$$

by arranging $\varepsilon_{m,n}$. A detailed Python code to generate $[\varepsilon]$ can be written as

```
1 import numpy as np
2
3 nFx = 2*M+1      # M: truncation order in x-direction
4 nFy = 2*N+1      # N: truncation order in y-direction
5 nSx = 4*M+1
6 nSy = 4*N+1
7
8 ind = np.arange(1, nFx*nFy+1, 1).reshape(1, nFx*nFy)
9 y = -((ind-1)%nFy) + nFy
10 x = np.int32((np.fliplr(ind)-y)/nFy + 1)
11 indm = x - x.T + nFx
12 indn = y - y.T + nFy
13 temp_ind = (indn-1)*nSx + indm
14 indmn = np.squeeze(temp_ind.reshape(1, nFx*nFy*nFx*
    nFy))
15
16 temp_eps = np.squeeze(epsmn.T.reshape(1, nSx*nSy))
17 # epsmn: Fourier transform of the permittivity
    matrix
18
19 BTTB = temp_eps[indmn-1].reshape(nFx*nFy, nFx*nFy)
```

where BTTB of the line 19 indicates $[\varepsilon]$.

4. Performance verification

In this section, we demonstrate computation tests using MAXIM to verify its calculation accuracy and appropriateness for various applications of dielectric metasurfaces. The calculation results are also compared with the commercial FDTD software

of Lumerical FDTD instead of other RCWA schemes. The computation scheme of FDTD is clearly distinct from that of RCWA, so we increase the credibility of the comparison and ensure the capability of MAXIM. To compare the results quantitatively, we use the root-mean-square deviation

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^{\rho} (\zeta_i - \xi_i)^2}{\rho \Omega^2}} \quad (42)$$

where ζ and ξ represent values of each data point calculated by MAXIM and FDTD respectively, Ω stands for the total range of the value, and ρ is the total number of data points. For example, $\Omega = 1$ for transmittance, reflectance and efficiency whereas $\Omega = 2\pi$ for phase.

As a result, we confirm that MAXIM is reliable simulation software for the investigation of dielectric metasurfaces. The results of MAXIM and FDTD differ by $< 1\%$ in all of the structures tested. Moreover, MAXIM inherently provides the complex transmission and reflection coefficients that are necessary for metasurface design, whereas the commercial FDTD software requires additional computation to obtain them. This feature drastically increases the user convenience of MAXIM. Considering those capabilities, MAXIM is competitive EM wave software for simulations in metasurfaces and nanophotonics.

4.1. Convergence test

The magnitude of the truncation orders M, N determines both the calculation accuracy and the computation costs in MAXIM. Simulations obtained using Fourier-series expansion of EM fields and permittivity distribution approach real values progressively as the truncation order increases; i.e., such orders are similar to the mesh resolution in FDTD and FEM. Increase in order usually yields increase in computation accuracy, but also increase in computation costs; a reasonable compromise should be reached.

The variation of computation results according to the truncation order is investigated using 1D and 2D rectangular silicon (Si) gratings on a silicon dioxide (SiO_2) substrate (Fig. 5). The 1D grating can be generated from a 2D rectangular structure by setting either M or $N = 0$. Despite of the numerical error of each method, the computation results of MAXIM differ from those of FDTD by $< 0.1\%$ for both 1D and 2D gratings. Given that the disparity $> 0.1\%$ even among RCWA schemes [18], MAXIM's solution can be considered highly reliable.

4.2. Fresnel equations

Transmission and reflection characteristics at the interface of two different media can be described using the Fresnel equations. In MAXIM this situation can be modeled by locating a zero-thickness film or no structure between input and output media. The Fresnel equations have exact solutions of complex transmission and reflection coefficients, so investigation of the dielectric interface is worthwhile to evaluate the accuracy of MAXIM.

In the Fresnel equations, the complex reflection coefficients are given by [38]

$$r_{\text{TE}} = \frac{n_1 \cos \theta_i - n_2 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i\right)^2}}{n_1 \cos \theta_i + n_2 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i\right)^2}} \quad (43)$$

$$r_{\text{TM}} = \frac{n_1 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i\right)^2} - n_2 \cos \theta_i}{n_1 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i\right)^2} + n_2 \cos \theta_i} \quad (44)$$

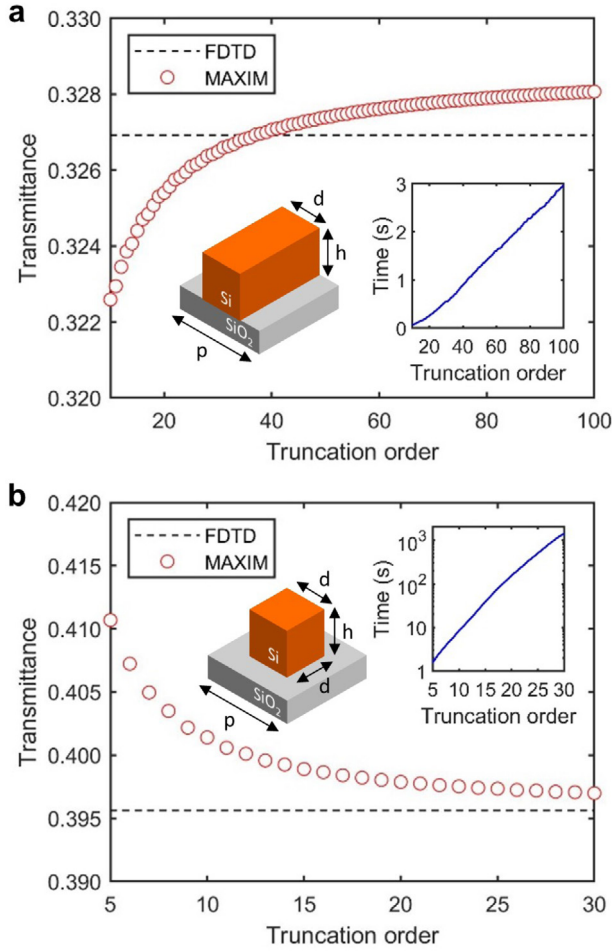


Fig. 5. Convergence test for transmittance. Linearly-polarized light of wavelength 514 nm is incident from a SiO_2 substrate to (a) 1D (p : 300 nm; h : 300 nm; d : 200 nm) and (b) 2D (p : 300 nm; h : 300 nm; d : 150 nm) Si grating structures. Discrepancies between MAXIM and FDTD decrease to $< 0.1\%$ as the truncation order increases. (inset) Measured computation time of MAXIM for each truncation order.

where TE and TM represent transverse electric and magnetic waves respectively, n_1 and n_2 indicate the refractive indices of input and output media, and θ_i is an incident angle. The reflectance of each wave at the interface can be obtained by calculating the absolute square of Eqs. (43) or (44).

The calculation result of the reflectance using MAXIM coincides with the reflectance calculated from the Fresnel equations throughout the incident angles from 0° to 89° (Fig. 6). For the case of $n_1 = 1$ and $n_2 = 2$, the total transmission of the TM wave occurs near the Brewster angle $\theta_B = \arctan(n_2/n_1) \approx 63.4^\circ$. For the opposite case of $n_1 = 2$ and $n_2 = 1$, θ_B becomes 26.6° , and total internal reflection of both TE and TM waves occurs at the critical angle $\theta_C = \arcsin(n_2/n_1) = 30^\circ$. The RMSD of those four cases is $< 10^{-15}$; this result confirms the reliability of MAXIM for this calculation.

4.3. Huygens metasurfaces

A Huygens metasurface is one of the most common frameworks in dielectric metasurfaces to manipulate light propagation. The basic unit structure of the Huygens metasurfaces has an inplane symmetry that allows operation insensitive to polarization. The name ‘‘Huygens’’ is originated from the famous Huygens–Fresnel principle in which every infinitesimal point of

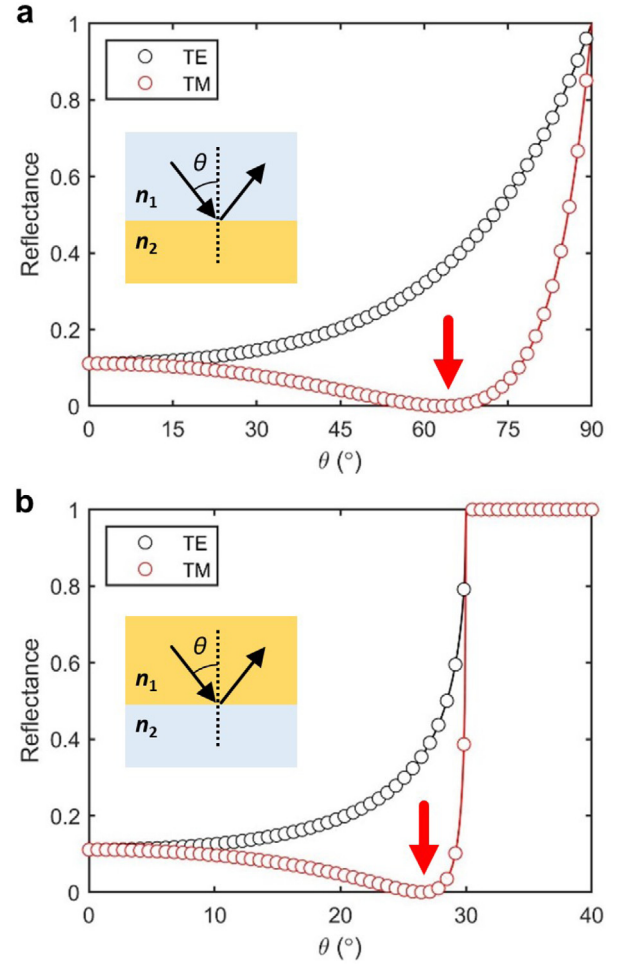


Fig. 6. Simulation of Fresnel equations. Linearly-polarized light is incident from the medium 1 that has refractive index n_1 to the medium 2 that has n_2 where (a) $n_1 = 1$, $n_2 = 2$ and (b) $n_1 = 2$, $n_2 = 1$. TE: transverse electric wave; TM: transverse magnetic wave. Real lines: analytic solutions calculated from the Fresnel equations; red arrows: at which total transmission occurs. The critical angles and Brewster angles predicted by MAXIM exactly correspond to the analytic solution.

a wavefront works as a point source of secondary wavelets. One critical assumption of the principle is that the secondary waves can propagate only in the forward direction. Similarly, the scattered light from the structures of Huygens metasurfaces travels practically in the forward direction, and yields high transmittance with low reflectance. This directive phenomenon can be described by the Kerker condition that elucidates the interference between the electric and magnetic dipole resonances that are induced within dielectric structures [12]. Moreover, keeping the Kerker condition, the transmission phase can be modulated from 0 to 2π by controlling the size of nanostructures [40], so the Huygens metasurface benefits from a high modulation efficiency in phase-tuning applications [41,42].

Cylindrical Si cavities are widely exploited in Huygens metasurfaces. MAXIM is used to characterize the transmission characteristics of such devices under the incidence of linearly polarized light (Fig. 7). As is usually done in metasurface research, the input medium of the substrate is set to SiO_2 . The calculation result clearly shows a high transmission range near the structure period of 175 nm. In this region, a full-phase excursion can be also achieved by varying the diameter of the cavity. The transmission loss is caused by the non-zero extinction coefficient of

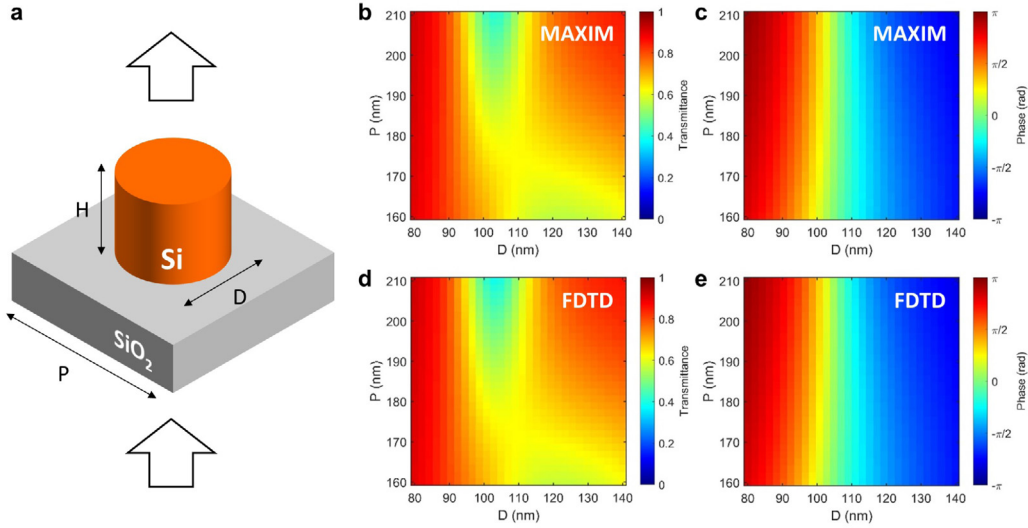


Fig. 7. Simulation of the Huygens metasurface. (a) Unit cell configuration consisting of a cylindrical Si cavity on a SiO₂ substrate. Linearly-polarized light of wavelength 532 nm is normally incident from the substrate to structure. H: 220 nm. The diameter D varies from 80 nm to 140 nm while the period P varies from 160 nm to 210 nm. Two arrows denote a light incidence and transmission direction. (b),(c) Calculated transmittance and phase retardation respectively using MAXIM. Truncation orders are set to 10 for both directions. (d),(e) Calculation results from FDTD.

Si at the wavelength of 532 nm. In the near-infrared wavelength regime, where the optical loss of Si becomes negligible, a Huygens metasurface can yield near-unity transmittance. However, such a low-loss material reduces the computation stability of FDTD, and thereby precludes the comparison with MAXIM; accordingly, our simulation is conducted in the visible regime despite the optical loss. The computation result of MAXIM corresponds closely to the result of FDTD, with RMSD = 0.0076 for transmittance and RMSE = 0.0046 for phase. Given that many of recent metasurfaces consist of low-loss dielectrics such as titanium dioxide and gallium nitride [2,43], MAXIM can be a promising simulation tool because the computation stability of RCWA is not affected by the transparency of materials.

4.4. Pancharatnam–Berry phase

The Pancharatnam–Berry phase, also called ‘geometric phase’ in optics, is another universal framework in metasurfaces that is used to control a spatial phase gradient of EM waves [44–46]. In contrast to the preceding Huygens metasurfaces that operate without regard to polarization, geometric metasurfaces work for the circularly polarized light incidence and outgoing waves are usually composed of co-polarization and cross-polarization components. Among them, only the cross-polarization component is affected by the geometric phase. Hence, the modulation efficiency of the geometric metasurface is simply the polarization conversion efficiency.

When left-handed circularly polarized light (LCP) strikes a rectangular structure, the Jones vector of the outgoing wave can be represented as [1]

$$\frac{\tilde{t}_l + \tilde{t}_s}{2} \begin{bmatrix} 1 \\ i \end{bmatrix} + \frac{\tilde{t}_l - \tilde{t}_s}{2} e^{i2\alpha} \begin{bmatrix} 1 \\ -i \end{bmatrix} \quad (45)$$

where $\tilde{t}_l$ and $\tilde{t}_s$ are complex transmission coefficients for the incident light linearly polarized along the long and short axes of the structure respectively, and α indicates an inplane rotation angle of the structure. The first term of Eq. (45) represents LCP; the second term indicates the cross-polarization component, which is right-handed circularly polarized light (RCP). Broadband operation is a major advantage of geometric metasurfaces, because the geometric phase of $e^{i2\alpha}$ is independent of wavelength.

Implementation of the geometric phase is highly intuitive compared with other frameworks; *i.e.*, a doubling of the rotation angle α is equal to the modulated phase. It is worth noting that the concept of optical geometric phase has existed before the emergence of metasurfaces [47]. However, recent progress on dielectric metasurfaces has drastically improved the capability of the geometric phase to enable achromatic operation [43] and arbitrary dispersion engineering [48].

The polarization conversion efficiency from the rectangular Si structure is characterized using MAXIM (Fig. 8). The conversion efficiency is defined as $|\tilde{t}_l - \tilde{t}_s|/2|^2$ that is the absolute square of the cross-polarization transmission coefficient in Eq. (45). The linearly polarized light is incident from the SiO₂ substrate to the structure. Two simulations are performed, one for each of two orthogonal polarizations, to obtain $\tilde{t}_l$ and $\tilde{t}_s$ in Eq. (45). Such a common simulation is also conducted using FDTD and the two results of the conversion efficiency agree closely with each other (RMSD = 0.0083); this confirms the applicability of MAXIM.

4.5. Structural color

Some dielectric metasurfaces can generate a broad range of colors that have high purity [49]. In dielectric metasurfaces that are composed of high-index subwavelength structures, the colors are produced as a result of sieving of incident EM waves by localized EM resonances. Therefore, transmission or reflection spectra from dielectric metasurfaces can be understood using multipole analysis of EM scattering. Associated unit structures are designed to tailor the multipole behavior, especially of electric and magnetic dipoles. So far, many kinds of structures, materials and their combinations have been studied to increase the color purity and functional capability for use in display applications [50].

The reflection spectrum of an exemplary structure [49] is calculated using MAXIM (Fig. 9). Linearly polarized light strikes the metasurface from the normal to the upper space. This structure features multilayered dielectrics to improve the color saturation compared with homogeneous dielectric structures. The top SiO₂ and bottom silicon nitride layers are designed to meet the index-matching condition that suppresses an electric quadrupole resonance occurring at non-resonant wavelengths. The calculated reflection spectrum clearly shows near-unity reflectance

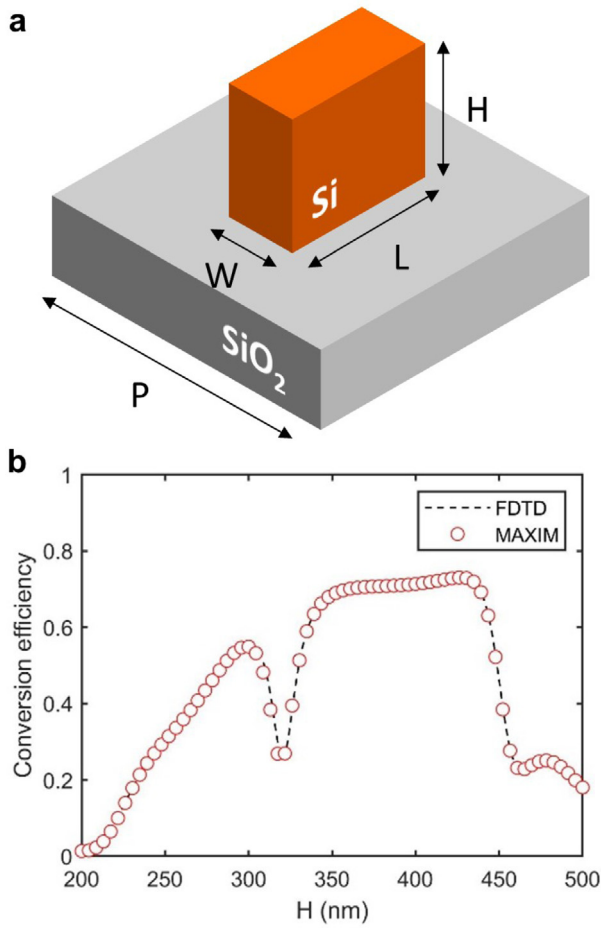


Fig. 8. Simulation of conversion efficiency of a geometric metasurface. (a) Unit cell configuration consisting of a rectangular Si structure on a SiO₂ substrate. Circularly-polarized light of wavelength 635 nm is normally incident from the substrate to structure. P: 350 nm, L: 190 nm, W: 100 nm. The structure height H varies from 200 nm to 500 nm. (b) Comparison of calculated polarization conversion efficiency. Truncation orders are set to 10 for both directions.

near 550 nm without any significant sub-peaks in the visible regime. The result of MAXIM agrees with the result of FDTD (RMSD = 0.0054); this result again verifies the validity of MAXIM simulations.

4.6. Interleaved metasurfaces

Multifunctionality is an advantage of metasurfaces over conventional refractive optical components [13,48]. Due to the nature of EM scattering from antennas, they can integrate different functions by wave parameters such as wavelength, polarization and incident angle [50]. An interleaved metasurface in which a unit cell contains multiple structures is a straightforward and effective framework to achieve this integration because fabrication of those metasurfaces does not require any additional processes compared to typical dielectric metasurfaces. Therefore, the interleaving method has been widely used to demonstrate many multifunctional metasurfaces [51]. The ability of MAXIM to simulate this kind of metasurface is also tested.

The representative structure of interleaved metasurfaces is examined using MAXIM (Fig. 10). The unit cell consists of four rectangular Si structures, of which two have the same dimension. The working principle is similar to the geometric phase, so each

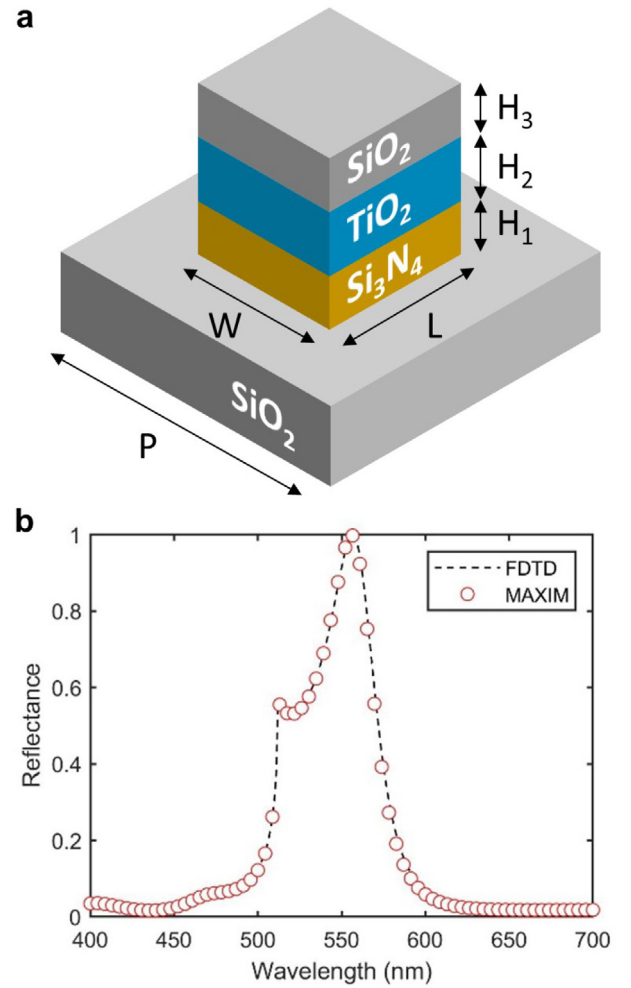


Fig. 9. Simulation of the reflectance spectrum from a color-generating metasurface. (a) Unit cell configuration consisting of multilayered rectangular dielectrics. Linearly-polarized light is normally incident from the air to structure. P: 350 nm; L: 200 nm; W: 200 nm; H₁: 60 nm; H₂: 140 nm; H₃: 100 nm. (b) Comparison of the calculated reflectance spectra in the visible regime. Truncation orders are set to 10 for both directions.

structure induces phase retardation of cross-polarized light by inplane rotation. The conversion efficiency is also defined as $|\langle \tilde{t}_l - \tilde{t}_s \rangle / 2|^2$ that is the absolute square of the cross-polarization transmission coefficient in Eq. (45). Each structure has an extremum of conversion efficiency in the vicinity of red, green or blue wavelengths by suppressing the efficiency at other wavelengths. Therefore, phase gradients of transmitted red, green and blue light can be manipulated independently by rotating the corresponding blocks. To encounter the effect of the superposition of all structures, the conversion efficiency of the unit cell is evaluated using MAXIM. Typical geometric metasurfaces that are composed of a single antenna usually have a single extremum of conversion efficiency in the visible regime. In contrast, the interleaved metasurface simulated using MAXIM clearly shows the multiple extrema induced by the multiple antennas, and thereby allows multiwavelength operation. The result of MAXIM also corresponds to the result of FDTD (RMSD = 0.0097). The discrepancy is a result of the index mismatch between MAXIM and FDTD, because the fitted index model of FDTD throughout the whole visible regime always deviates from the input data that are used in MAXIM.

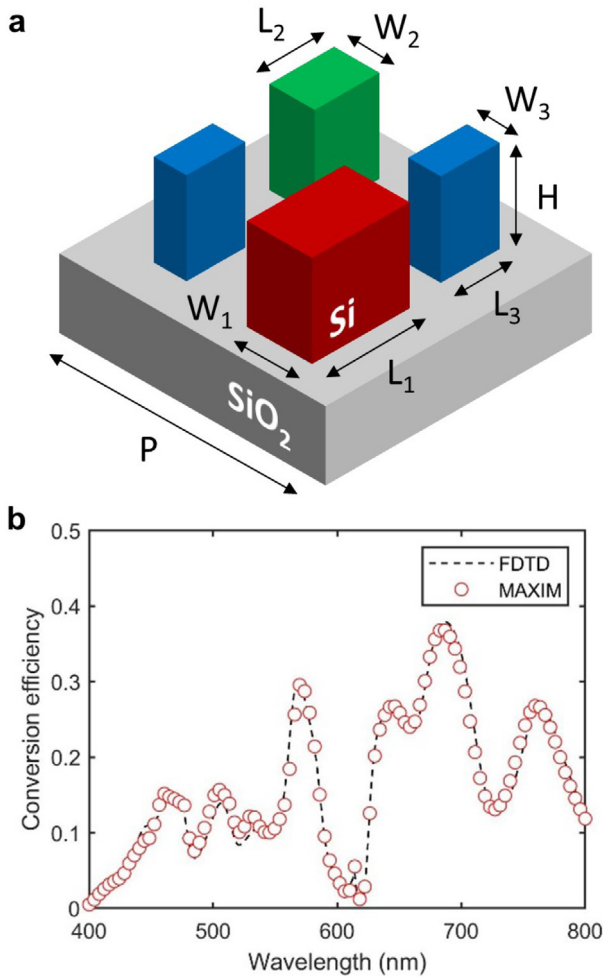


Fig. 10. Simulation of conversion efficiency from an interleaved metasurface. (a) Unit cell configuration composed of four rectangular Si structures, of which two have the same dimensions. Circularly-polarized light is normally incident from the substrate to structures. Color of each structure corresponds to its dimension. P: 420 nm; H: 320 nm; L_1 : 155 nm; W_1 : 110 nm; L_2 : 100 nm; W_2 : 70 nm; L_3 : 80 nm; W_3 : 35 nm. Centers of structures are located at the centers of each quadrant of the square unit cell. (b) Comparison of calculated polarization conversion efficiency in the visible regime. Truncation orders are set to 15 for both directions.

4.7. Multilayered structures

Stacking of metasurfaces is another standard method to achieve multifunctionality by increasing the DoF in structural configuration [52]. The working principle of the multilayered metasurface can be described as a superposition of the effect of individual structures in the Fresnel region without interaction among stacked metasurfaces. This method benefits from relaxed structural confinement, so the structure dimension can vary within the whole unit cell space whereas the available space of each structure in previous interleaved metasurfaces is highly restricted by other components. Moreover, the multilayered metasurface is still much thinner than conventional optical elements, so this type of metasurfaces is favorable for use in ultrathin multifunctional devices by integrating individual metasurfaces.

The transmission properties of two layers of Huygens metasurfaces stacked on each other in the presence of a SiO_2 interlayer are investigated using MAXIM (Fig. 11). Both metasurfaces consist of cylindrical Si cavities of various diameters. The bottom layer

is defined on a SiO_2 substrate, and the medium surrounding the lower cavity is also filled with SiO_2 . A 1- μm -thick SiO_2 interlayer and another layer of cylindrical Si cavities is stacked on the bottom layer in sequence. The surrounding medium of the top Si cavity is filled with air, and linearly polarized light is incident from the substrate to the structures. The calculation result verifies that increasing the DoF is an effective way to tailor the transmission coefficients, compared with single-layered metasurfaces. The results agree with FDTD results (RMSD = 0.005 for transmittance; RMSD = 0.0037 for transmission phase). This agreement further confirms the validity of MAXIM simulations.

4.8. Chiral metasurfaces

Chirality that is a geometrical property of an image which cannot coincide with its mirror image is abundant in biological materials and molecules such as DNA, proteins and amino acids. Those materials induce chiroptical effects of optical activity and circular dichroism (CD). Detection and measurements of such chiral properties using other chiral materials have fundamental applications in biochemistry and the pharmaceutical industry [53]. Natural chiral materials can be used for the purpose, but usually suffer from low chiral responses requiring a long optical path to achieve conspicuous discrepancy between incident LCP and RCP. In contrast, dielectric metasurfaces thinner than the operating wavelength can provide a promising solution to drastically increase the chiral response due to high EM field enhancement and low optical loss within the structures [54].

The CD of a Z-shaped Si antenna is calculated using MAXIM in the visible regime (Fig. 12). Calculation of CD requires the complex transmission coefficients of a circular polarization base. The base conversion relation from a linear polarization to a circular polarization is represented as [55]

$$\begin{bmatrix} t_{ll} & t_{lr} \\ t_{rl} & t_{rr} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} t_{xx} + t_{yy} + j(t_{yx} - t_{xy}) & t_{xx} - t_{yy} - j(t_{yx} + t_{xy}) \\ t_{xx} - t_{yy} + j(t_{yx} + t_{xy}) & t_{xx} + t_{yy} - j(t_{yx} - t_{xy}) \end{bmatrix} \quad (46)$$

where t_{ij} is a complex transmission coefficient, its first and second subscriptions indicate the polarization states of transmitted and incident light, x and y denote a linear polarization along the x and y direction, l and r stand for LCP and RCP, respectively. The total transmittance for the incident LCP and RCP are represented as $T_l = |t_{ll}|^2 + |t_{rl}|^2$ and $T_r = |t_{rr}|^2 + |t_{lr}|^2$, respectively, and the CD is defined as $\Delta T = T_l - T_r$. MAXIM provides full information about the complex transmission coefficients under linearly-polarized light incidence, so the CD of the Z-shaped antenna can be obtained without difficulty. The calculated CD oscillates rapidly in the visible regime because the dimension of the structure is chosen randomly without optimization. Despite the violent oscillation in the CD plot, the result of MAXIM agrees well with the result of FDTD (RMSD = 0.0089). This result further validates the capability of MAXIM.

4.9. Further development

In this section, we verified the computation accuracy of MAXIM compared with the commercial FDTD software. The discrepancies in each example are evaluated quantitatively and RMSDs calculated for the structures tested in this work are all < 0.01 ; this agreement confirms that the calculation result of MAXIM for dielectric metasurfaces coincides with the result of FDTD within 1% difference, and verifies that MAXIM is suitable for practical use.

The capability of MAXIM can be further improved by incorporating a mathematical treatment to facilitate fast convergence

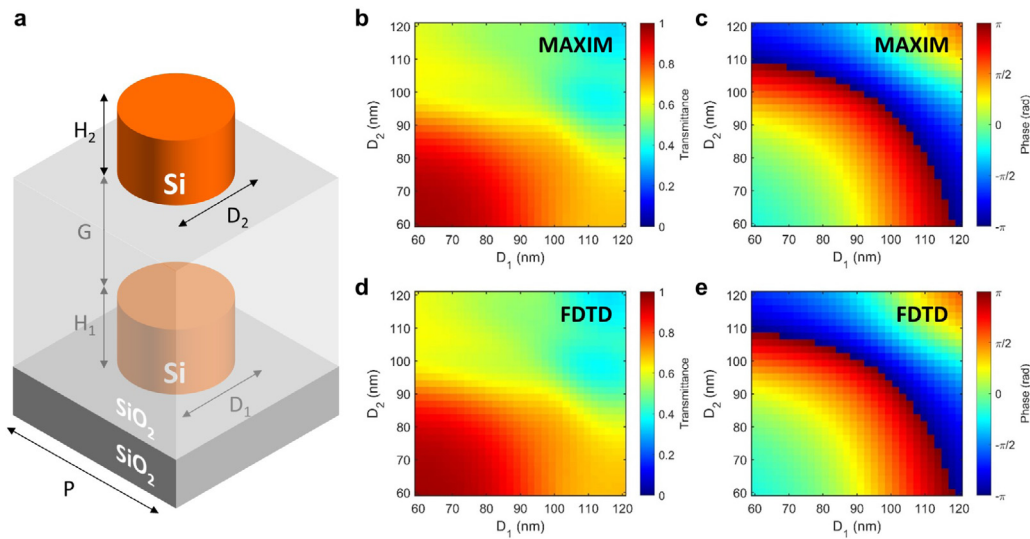


Fig. 11. Simulation of the multilayered metasurface. (a) Unit cell configuration consisting of two cylindrical cavities with an intermediate layer of SiO₂. The bottom cavity is also surrounded by SiO₂. Linearly-polarized light of wavelength 532 nm is normally incident from the substrate to structures. P: 175 nm; H₁: 220 nm; H₂: 220 nm; G: 1 μm. Diameters D₁ and D₂ of the top cavity vary from 60 nm to 120 nm. (b),(c) Calculated transmittance and phase retardation using MAXIM, respectively. Truncation orders are set to 10 for both directions. (d),(e) Calculation results from FDTD.

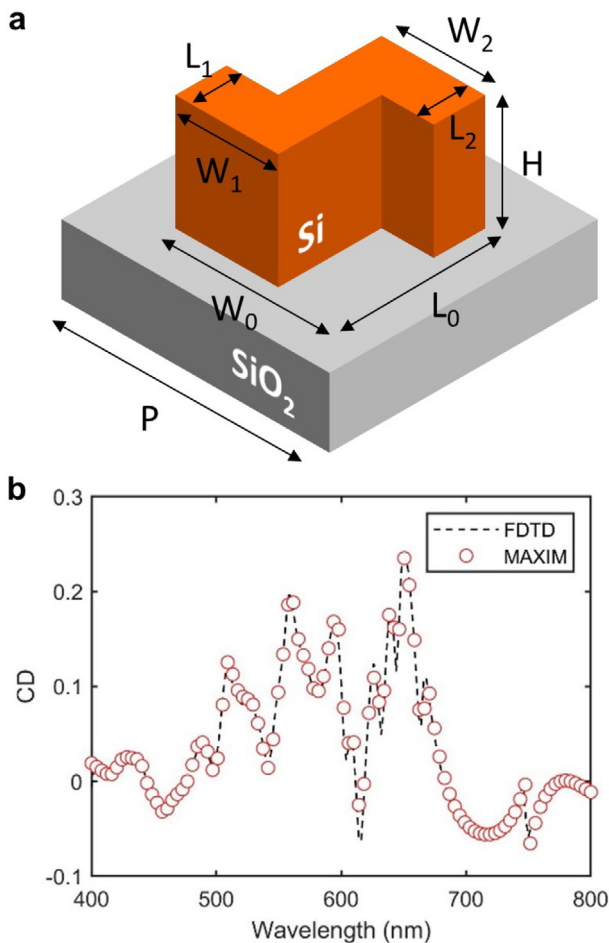


Fig. 12. Simulation of a chiral metasurface. (a) Unit cell configuration consisting of a Z-shaped Si structure. Circularly-polarized light is normally incident from the substrate to structures. P: 400 nm; H: 300 nm; L₀: 300 nm; L₁: 100 nm; L₂: 100 nm; W₀: 300 nm; W₁: 200 nm; W₂: 200 nm. (b) Comparison of calculated circular dichroism (CD) in the visible regime. Truncation orders are set to 20 for both directions.

for TM waves. Although most of typical metasurfaces operate for normally incident plane waves regardless of polarization mode, the poor convergence for TM waves has been a long-standing problem in RCWA. Many methods have been also developed and investigated to improve this convergence drastically [56–60]. The most famous treatment is perhaps to use a BTTB of the reciprocal permittivity $[[1/\epsilon]]^{-1}$ instead of the ordinary BTTB $[[\epsilon]]$ in Eq. (7) [57]. In the limit of infinite truncation orders, $[[1/\epsilon]]^{-1}$ is identical to $[[\epsilon]]$ [28]. The theoretical elucidation of using $[[1/\epsilon]]^{-1}$ has been also presented with a correct usage, called Li's factorization rules which specify a criterion by considering the continuity of the electric field and electric displacement [36]. These rules can drastically improve the convergence of RCWA for TM waves. The only problem is to estimate electric field components perpendicular and parallel to structure boundaries. The electric field can be easily decomposed in 1D gratings, but the problem becomes complicated in 2D gratings. Therefore, a normal vector method has been developed and implemented to facilitate the decomposition in 2D gratings [28,61]. Besides, it is also possible to isolate a structure using perfectly matched layers which increase the availability of RCWA remarkably [62]. These fascinating techniques will be incorporated in subsequent version of MAXIM.

5. Conclusion

We present MAXIM, which is EM wave simulation software that uses RCWA. A highly user-friendly GUI is a primary advantage of MAXIM compared with conventional open-source RCWA programs, which usually suffer from low accessibility, especially for beginners who are not familiar with computer programming. The table-based intuitive GUI of MAXIM is extremely easy and straightforward, so MAXIM can provide a real opportunity to introduce RCWA to the communities of related fields.

The basic formulation of MAXIM is conventional RCWA associated with ESMM to allow computation of total coupling coefficients in a similar manner to that of a total S-matrix. MAXIM also features high DoF in structural configuration covering almost arbitrary shapes for 2D gratings. The main application of MAXIM is focused on dielectric metasurfaces and the computation accuracy of MAXIM is investigated by applying it to several didactic

examples. All of MAXIM's results are within 1% of that obtained using commercial FDTD software. Due to those easy accessibility, wide availability and high reliability, MAXIM will be a practical RCWA platform in the fields of metasurfaces and nanophotonics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was financially supported by the National Research Foundation (NRF), Republic of Korea grants (NRF-2019R1A2C3003129, CAMM-2019M3A6B3030637, NRF-2019R1A5A8080290, NRF-2018M3D1A1058997) funded by the Ministry of Science and ICT of the Korean government.

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